Title

Fast Geometric Machine Learning Methods for Brain Image Analysis

Abstract

Alzheimer's Disease (AD), the most common form of dementia, is characterised by the accumulation of misfolded proteins in the brain, ultimately leading to cortical atrophy and cognitive decline. Accurate measurement of the atrophy patterns is essential not only for diagnosis but also for early detection and potential prevention (Lenhart et al., 2021).

However, due to the high variability in cortical geometry across individuals, population-based analyses typically rely on long, sophisticated neuroimaging pipelines for cortical feature computation and inter-subject alignment (Dale et al., 1999; Fischl, 2012; Fischl et al., 1999). These traditional approaches often require several days to weeks of computation for large cohorts, creating a major bottleneck for large-scale neuroimaging studies. Deep learning–based approaches have the potential to dramatically accelerate this process (Ma et al., 2025).

In this talk, I will present the recent work conducted in our team to streamline the cortical surface-based pipeline. I will review optimization-based approaches and end-to-end deep learning methods for cortical mesh reconstruction and surface registration. Some insights into the Spherical Demons algorithm (Yeo et al., 2010), a classical diffeomorphic registration method on the sphere, will be provided. I will then introduce NC-Reg, our novel approach for rigid registration of spherical surfaces using a neural representation. Subsequently, I will outline our ongoing and future work, with a particular focus on CorticalFlow++ (Cruz et al., 2022; Lebrat et al., 2021), a deep learning model for fast cortical mesh reconstruction directly from MRI images.

We believe that this work lays a strong foundation for more efficient, scalable, and robust tools for cortical surface analysis. These developments have the potential to enable large-cohort neuroimaging studies and ultimately contribute to both computational neuroscience and clinical applications.

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